

AI-Powered Food Loss Prediction & Redistribution System

From Loss to Logic: AI-Driven Food Redistribution for Indonesia and ASEAN

Team Dollar Warrior
Sam Ratulangi University



Natalio Michael Tumuahi



Christian Andreas Roeroe

Executive Summary



- 13.3% of food is lost post-harvest before reaching the retail stage.
- Indonesia's horticultural loss rate exceeds 60%, more than 2x the global average.
- These losses cost Indonesia an estimated Rp 213–551 trillion in annual economic losses.



A multi-layered AI system combining time-series forecasting, YOLOv8 computer vision, and OCR.



The pilot study at Bersehati Market targets a moderate 20% loss reduction rate. This is projected to save 10.2 tons of food and prevent approximately Rp 584 million in economic losses annually from a single market location.

Overview

13.3%

Global post-harvest food loss

25.4%

Global Loss rate for fruits & vegetables (highest category)

>60%

Indonesia's vegetable loss and waste exceeds 60%, more than 2x the global fruit & vegetable loss rate

Three core issues



Weather

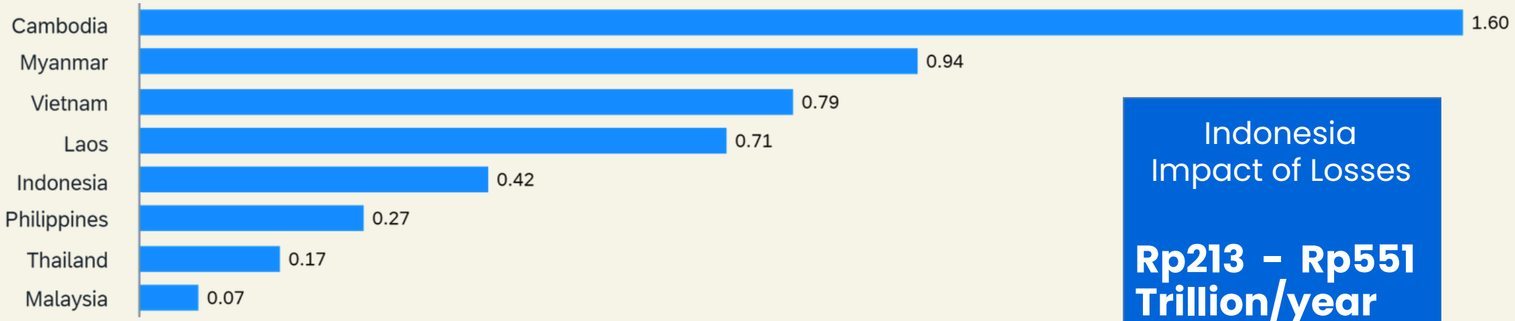


Data Fragmentation



Manual Data Entry

Food Loss Rankings in ASEAN (% of National GDP)



Indonesia
Impact of Losses

Rp213 - Rp551 Trillion/year

Aligned with **SDG 12.3** and **ASEAN AIFS Strategic Thrust 3**, we focus on **Bersehati & Pinasungkulan markets, Manado**, targeting **red bird's eye chili and tomatoes**, whose extreme price volatility from inter-provincial supply dependency makes them a clear case study of Indonesia's distribution-stage food loss crisis.

References: Bappenas (2021), FAO (2022), Jaroebsathapornkul, J. (2021), IPB University (2026), FAO (2026), ASEAN (2020) , Eriyanto et al.(2024).

Strategic Alignment

Layer	Framework	Specific reference	Contribute
Global	SDG 12.3	Indicator 12.3.1(a) Food Loss Index (production up to, not including, retail)	Early warning and redistribution at the distribution stage
	SDG 2.4	Resilient food production systems	Data-driven stock decisions for small vendors
	SDG 13.3	Early warning and adaptive capacity	Weather and price signals in the risk score
ASEAN food	AIFS Framework, SPA-FS Thrust 3	Integrated food security information systems to forecast, plan and monitor supplies	A field-level prototype of regional early warning
	FAF-SP 2026-2030	ST3 (Action Programmes 3-4), ST4 (AP 2-3), ST5 (AP 1-2)	AI, digital platforms, reduced post-harvest loss, PPP
ASEAN Community	AEC 2026-2030	Obj. 3.1 digital transformation; 3.9 MSMEs; 5.1 resilient supply chain	Zero-cold-chain tools for micro traders; digitalised supply data
	ASCC 2026-2030	Measure 1.1.6 digital agriculture; 6.1.6 food price shocks; Goal 2 inclusion	Protects vulnerable vendors and surplus reaches food-insecure groups

References: UN SDG Metadata; ASEAN (2020) AIFS-SPA-FS; ASEAN FAF-SP 2026-2030; AEC and ASCC Strategic Plans 2026-2030.

Problem Statement

“Broken since 2016 (Pinasungkulan)”
“Zero Infrastructure (Bersehati)”



Saved Commodities & Empowered Vendors



How might we empower traditional market distributors with a real-time, zero infrastructure predictive system to anticipate oversupply and intelligently redistribute highly perishable crops before physical degradation occurs?

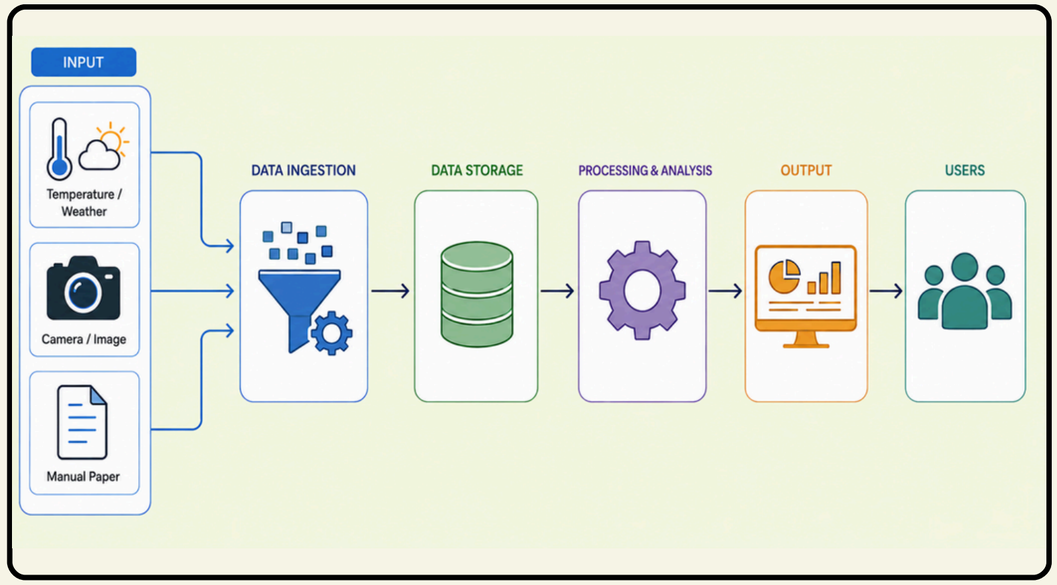
References: Traditional Market Vendors, Picture generated with ChatGPT.

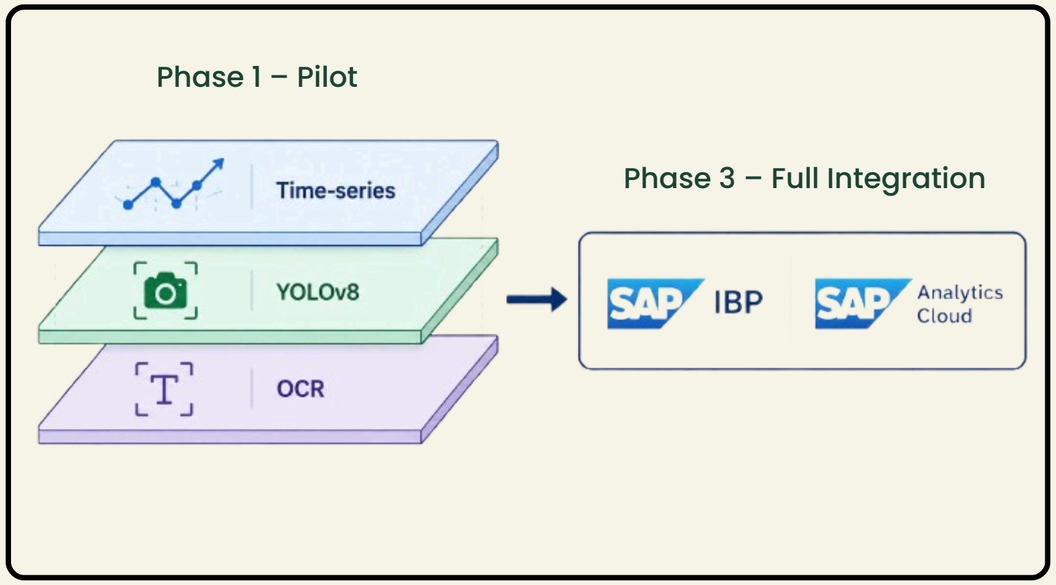
Current Initiatives

Title	Description	Strengths	Weakness
Bio Loop - From Waste to Worth (Team Mamamamoodeng, 2025)	A logistics app that processes organic waste into animal feed using Black Soldier Fly larvae	A circular economy solution integrated with IoT tracking	Curative in nature (food must first rot); diverts food nutrients from human consumption to animal feed.
Cut The Waste (Untitle Team, 2023)	An app for tracking expiration dates and household kitchen inventory	Features for using up leftovers and practical alerts to reduce waste at the consumer level	Relies entirely on manual data entry by users; does not address food loss at the upstream end of the distribution chain
Waste To Worth: Driving Sustainable Food System in ASEAN Countries Through Food Waste Reduction (Little EinScience, 2023)	An incentive based app collects household food waste to convert it into biofuel	High public participation through gamification	Addresses food that has already become waste, rather than preventing spoilage early in the supply chain; infrastructure cost ~ USD 2.4 million

References: ASEAN Data Science Explorer Storyboard.

Solution Overview

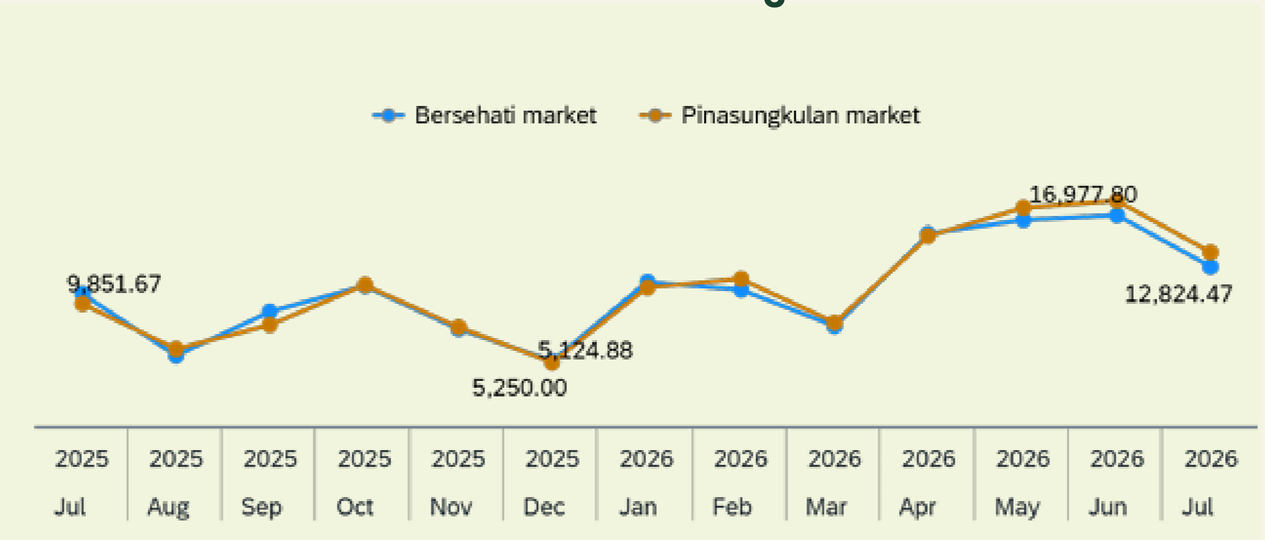




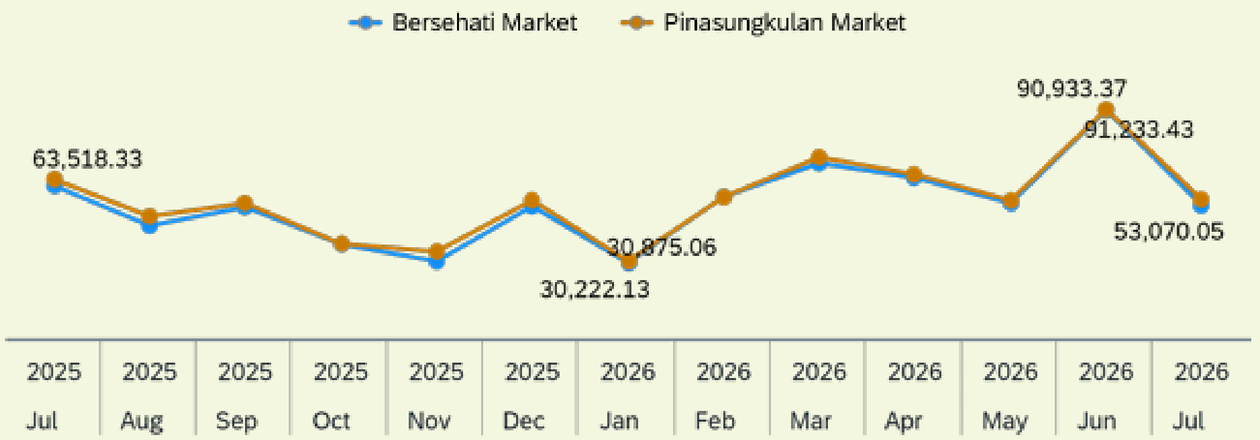
- We bypass the limitations of single-model approaches by combining three complementary AI layers:
  1. **Time-Series Forecasting:** Generates early warning risk scores based on weather anomalies, historical price trends, and volatility.
  2. **Computer Vision (YOLOv8):** Analyzes warehouse camera feeds to classify the actual visual condition of commodities, designed to calibrate the time-series risk score.
  3. **Optical Character Recognition (OCR):** Digitizes manual, paper-based stock records to seamlessly bridge traditional markets into the digital ecosystem.
  4. **Redistribution Trigger:** Critical risk alerts from RicaWatch prompt Perumda Pasar Manado to coordinate sameday surplus pickup with Surplus Indonesia/SOS in a fully automated via API in Phase 2.
- During the pilot phase, data was processed using Python (Google Colab), with SAP Analytics Cloud (SAC) used to generate the visualizations presented in this storyboard; full-scale integration with SAP IBP follows in Phase 3

Price Trends: Chili vs Tomato

Average Price Comparison of Tomato: Bersehati vs. Pinasungkulan



Average Price Comparison of Chili Pepper Bersehati vs. Pinasungkulan



Key Insight

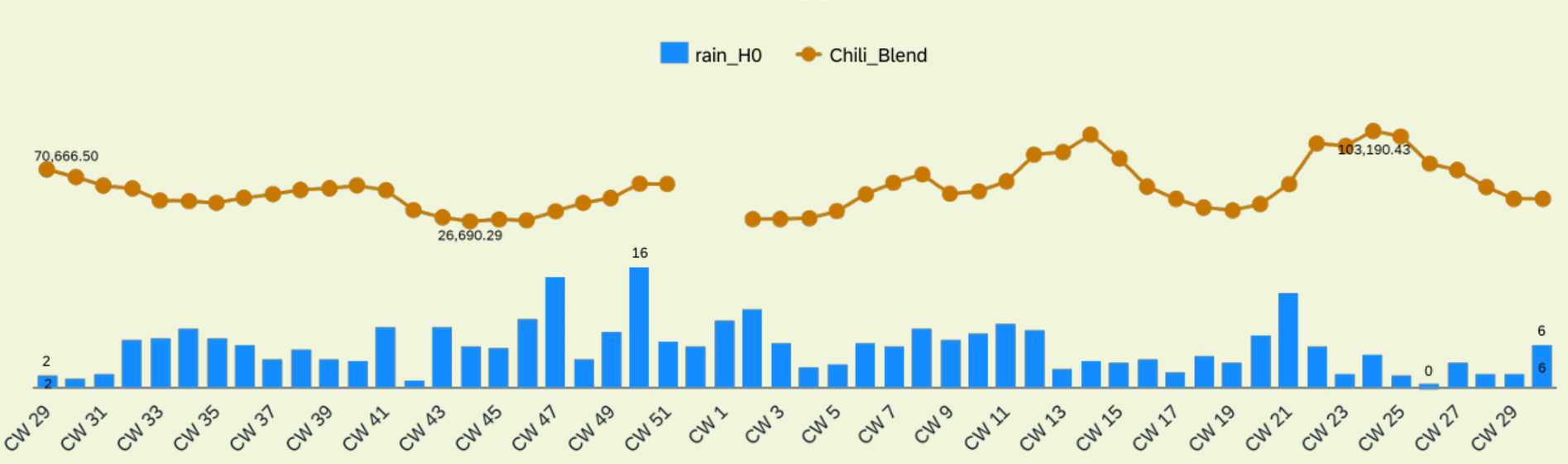
The lowest price points for chili peppers (January 2026: ~IDR 30,500/kg) and tomatoes (December 2025: ~IDR 5,190/kg) differed by one month. However, both peaked simultaneously in June 2026 (chili peppers ~IDR 90,900–91,200/kg; tomatoes ~IDR 17,800–18,100/kg) due to regional seasonal factors that concurrently constrained supply.

References: North Sulawesi Food Security Agency (Dinas Ketahanan Pangan Provinsi Sulawesi Utara), daily commodity price posts on Facebook @dishanpansulut, Jul 2025 - Jul 2026; manually compiled by Team Dollar Warrior.

Note: Chart generated using SAP Analytics Cloud.

Does Weather Explain Price Volatility? Not Alone?

Gorontalo Case Study: Rainfall (H0) vs. Aggregate Chili Price per Calendar Week



Data Limitation
CW52 2025 and CW1 2026 have no price data. They are excluded, not treated as zero (null ≠ 0)

So What?
Early warning must combine price trends, supplier-region weather, volatility, and visual condition

Key Insight

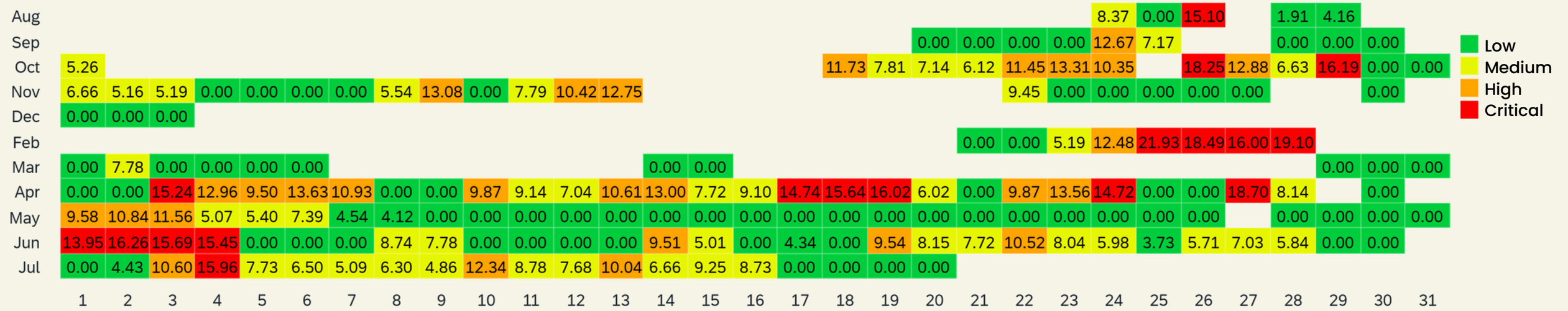
- Gorontalo is the dominant chili supply source for both markets, supporting its selection as the primary rainfall case study.
- Rainfall peaked at ~14.9–16.2 mm/week (CW47–CW50), while chili prices remained moderate–low (~Rp35,262–Rp58,556/kg). Correlation is weak (r = -0.268, p = 0.054) and only 1 of 20 weather price pairs is statistically significant (see Appendix)
- Annual price peaks (~Rp100,143 in CW14 and ~Rp103,190/kg in CW24) occurred 1–2 weeks after Idul Fitri and Idul Adha, while rainfall remained low (2.6–4.4 mm), pointing to post-holiday demand as a stronger driver.
- Traders reported mixed weather effects; secondary suppliers likely absorb local rainfall shocks.

References: North Sulawesi Food Security Agency, daily commodity price posts on Facebook @dishanpansulut, Jul 2025 - Jul 2026, manually compiled by Team Dollar Warrior; NASA POWER (weather data).

Note: Chart generated using SAP Analytics Cloud.

Oversupply Risk Calendar

Oversupply Risk Map for Chili Peppers Daily Calendar (July 2025 – July 2026)



Key Insight

- Both Critical-day clusters were driven by sharp price corrections, not rainfall: chili prices fell 43% (Rp71,667 to Rp41,167/kg), while rainfall on the lowest-price day was only 2.47 mm.
- Perumda Pasar Manado linked these corrections to simultaneous harvests across multiple production centers, making oversupply a stronger driver than same-week rainfall.
- The weak Gorontalo rainfall–price correlation ($r = -0.268$, $p = 0.054$) further indicates that weather alone has limited explanatory power for short-term price swings.
- Despite the Critical spikes, the highest risk score was only 22 on the observed scale, consistent with small daily spoilage volumes and traders' practice of holding unsold stock, indicating moderate-scale oversupply.

References: North Sulawesi Food Security Agency, daily commodity price posts on Facebook @dishanpansulut, Jul 2025 - Jul 2026, manually compiled by Team Dollar Warrior; NASA POWER (weather data); interview with Perumda Pasar Manado, Aug 2026.

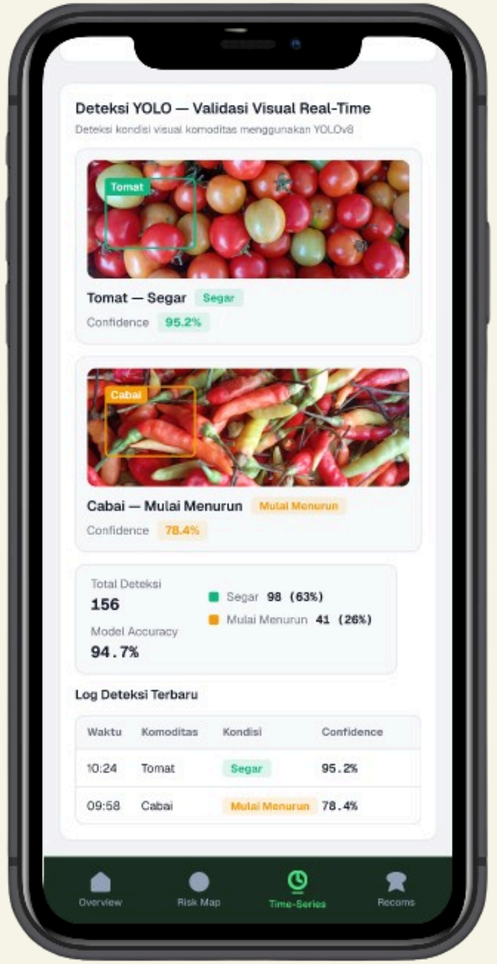
Note: Chart generated using SAP Analytics Cloud

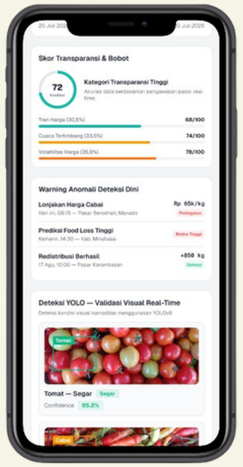
RicaWatch

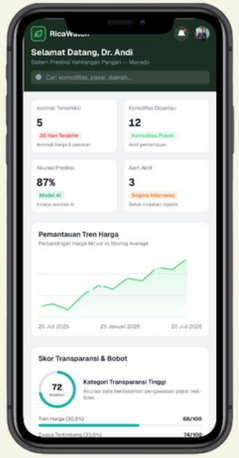
RicaWatch is a multi-role platform that customize data display and depth according to user needs;

- **Perumda Pusat Pasar Manado** - Comprehensive visibility across all markets in the City of Manado: regional risk maps, cross-location price trends, and supply allocation recommendations for strategic decision-making.
- **Market Unit Managers** - Data filtered according to the markets under their responsibility, for daily operational monitoring and coordination with traders.
- **Vendors** - Personalized, easy-to-understand risk alerts for the commodities they trade, without the complexity of irrelevant aggregate data.

Note: This screen is the interface for Perumda Pusat Pasar Manado. For the manager and vendor interfaces are in the Appendix. All figures shown are illustrative sample data.



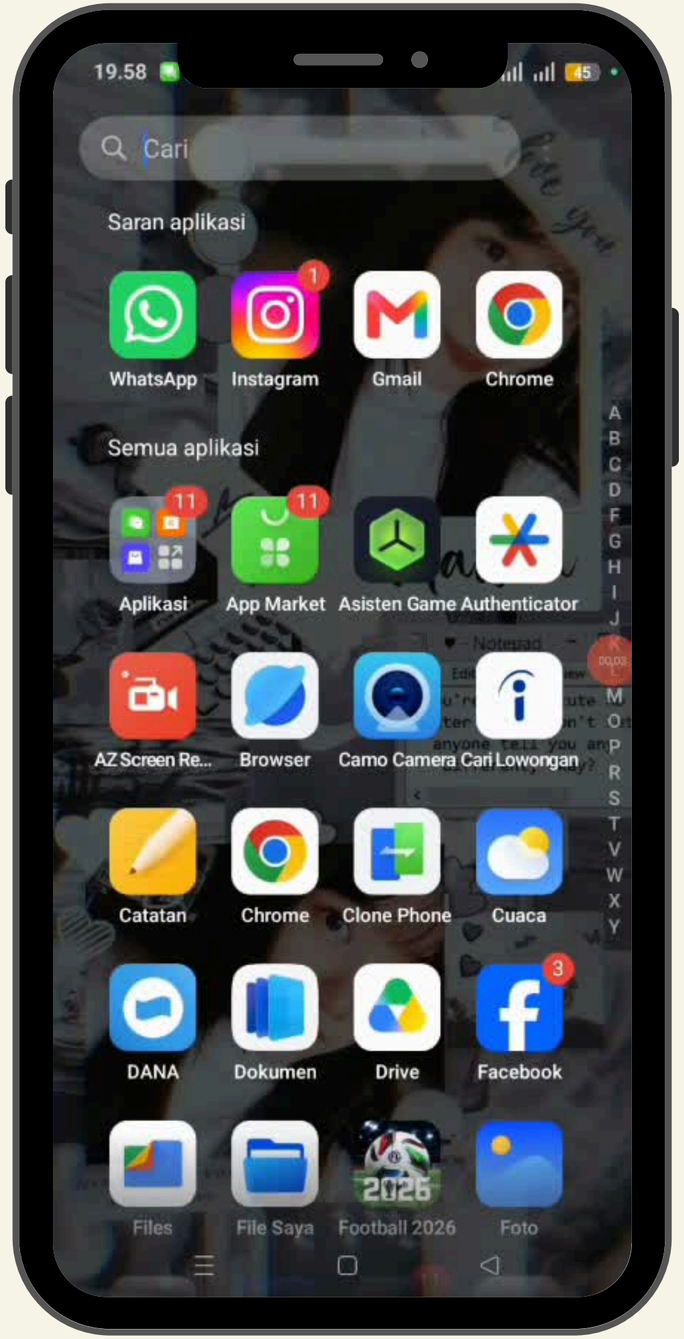




References: Team Dollar Warrior, Application Interface made by Figma.

App Demo

YOLOv8 proof of concept classifying tomato and chili freshness. Real model output, not mockup.



References: Prototype Application made by Team Dollar Warrior.

Impact

Target Group



Traditional Market Vendors



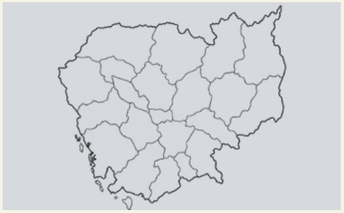
Smallholder farmers



Food insecure populations

Cross-ASEAN Scalability

Cambodia



- Worst macro food loss in ASEAN: 1.60% of GDP
- Fruits & vegetables loss: 20–45%, driven by non-refrigerated transport and traditional packaging

Myanmar



- 2nd worst macro food loss in ASEAN: 0.94% of GDP
- Fresh vegetables loss: 30–35%, worsened by power outages disabling cold storage

AIFS Framework Collaboration

Impact Scale

National Estimate

Rp 10–20 Trillion/year

A 10% reductionin losses at the distribution stage

Quantitative Estimation with a Case Study on Bersehati

VLB = 51.2 ton/year

VS (R = 20%) = 10.2 ton/year

Economic Value Estimation = 584 M / year

Inclusivity

- Current losses are borne entirely by individual merchants; the return mechanism to suppliers merely shifts the losses upstream it does not eliminate them.
- The solution provides decision making tools without requiring investment in relevant infrastructure, as cold storage is not operational in either pilot market (Pinasungkulan since 2016; Bersehati has never had it for horticultural products).
- • Targets small traders in traditional markets, not just large-scale supply chain players.

Note: The calculations used to estimate the economic value are provided in the appendix.

References: Jaroebsathapornkul, J. (2021), FAO (2016), ASEAN (2020), Images: Team Dollar Warrior, Pixabay (Cuong_art, geralt).

Stakeholders Roles

Direct Field Stakeholders & Data Providers

Stakeholders	Role	Engagement Status
Traditional Market Vendors	End-users of predictive tool; primary source of real-world food loss data	Validated (14+ Vendor Interviews)
Perumda Pasar Manado	Pilot location manager; operational supervisor & field facilitator	Validated
North Sulawesi Food Security Agency (BMKG planned)	Daily price benchmarks; weather via NASA POWER (BMKG data planned)	Integrated (price data); BMKG planned

Scaling & Implementation Partners

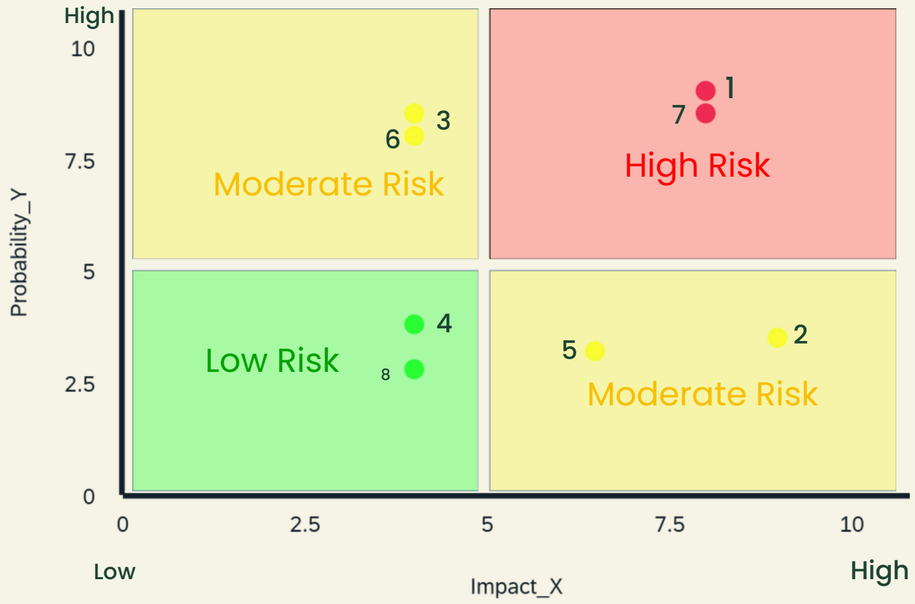
Stakeholders	Partnership Focus	Target Implementation Role
National Food Agency (Bapanas), Bulog & Coordinating Ministry for Food	Policy Framework & National Scale	National policy backing, data-sharing MoU, and execution of large-scale regional food redistribution
Surplus Food Networks (Surplus Indonesia, SOS, Foodbank of Indonesia)	Social Redistribution Channels	Off-taking edible oversupply crops and distributing them to food-insecure populations
Indonesian Cold Chain Association (ARPI)	Infrastructure Technical Partner	Long-term technical advocacy and cold storage rehabilitation

Note: Validated through 14+ vendor interviews and meetings with Perumda Pasar Manado.

References: Team Dollar Warrior.

Risks & Costs

Risk Assessment: Probability vs. Impact



1. Cold storage is not functioning	5. Low technology adoption
2. Weak correlation between weather and prices	6. Calculation assumptions have not been validated
3. Inconsistent inventory records	7. Complexity of multi-source supply
4. Slow bureaucratic response	8. Historical data is unstructured

Cost Summary

Implementation Phase	Low-End Scenario	High-End Scenario
Phase 1: AI Pilot	CAPEX: Rp 28.2 Million, OPEX/year: Rp 13.0 million	CAPEX: Rp70.7 million, OPEX/year: Rp 34.3 million
Phase 2: IoT Expansion	CAPEX: Rp12,0 Million	CAPEX: Rp25.0 Million (Estimated)

Note: Full details on technical specifications and unit prices for components (such as IP CCTV, NVIDIA Jetson, and ESP32 sensor nodes) are available in the Appendix.

References: Team Dollar Warrior (field interviews, Aug 2026; market price estimates).

Note: Chart generated using SAP Analytics Cloud.

Roadmap

Quarter 4 2026



PHASE 1
PILOT

- Location: Bersehati & Pinasungkulan Karombasan Markets, Manado.
- Commodities: Bird's eye chili peppers & tomatoes.
- Baseline data: 12 months of price data, multi-location weather data, established, field connections (August 2026).
- Deliverables: YOLOv8 proof of concept (1 market location) + OCR (sample manual documents).

2027 - 2028



PHASE 2
EXPANSION

- Expansion to various commodities (with a focus on products with a short shelf life: vegetables and fruits).
- Expansion to various provinces.
- Research on low-cost IoT temperature and humidity sensors (in response to the finding that there is a lack of cold storage facilities).
- Collaboration between ARPI and organizations that distribute surplus crops.

2028 - 2032



PHASE 3
FULL INTEGRATION

- SAP IBP + SAC integration at the provincial/national level.
- Memorandum of Understanding (MoU) on data sharing with Bappenas and Bulog.
- Replication to other provinces in Indonesia.
- Potential for replication in ASEAN (Cambodia, Myanmar) via the AIFS Framework.

Pilot Phase 1 launches via a seed grant/CSR-funded proof-of-concept (limited scope: 1 camera + sample OCR), with full CAPEX secured through a follow-on proposal to Perumda Pasar Manado; Phase 2 OPEX transitions to Manado City APBD via Perumda; Phase 3 scales to national APBN funding through Bapanas MoU.

References: **Images:** Team Dollar Warrior, Pixabay (geralt).

Conclusion

1. **Problem:** Manado's traditional markets lose crops every day due to a lack of cold-chain infrastructure and fragmented data.
2. **Solution:** A 3-layer AI system (Time-Series + YOLOv8 + OCR) requiring zero infrastructure, designed for zero-infrastructure deployment; pilot-ready
3. **Impact:** 10.2 metric tons of food & ~Rp584 million/year saved in 1 market (moderate scenario); potential of Rp10–20 trillion/year nationwide.
4. **Scalability:** Designed for replication in Cambodia, Myanmar, and other ASEAN countries through the AIFS framework.
5. **Roadmap:** Pilot in Q4 2026 → Expansion in 2027–2028 → Full integration with SAP IBP/SAC in 2028–2032.

An affordable solution, based on real-world field data, and ready to scale. Turning the food loss crisis into an opportunity for ASEAN food security.

THANK YOU

References

ASEAN. (2020). *ASEAN integrated food security (AIFS) framework and strategic plan of action on food security in the ASEAN region (SPA-FS) 2021–2025*. https://asean.org/asean-integrated-food-security-aifs-framework-and-strategic-plan-of-action-on-food-security-spa-fs-2021-2025-2/

ASEAN. (2025a). *ASEAN Economic Community strategic plan 2026–2030*. ASEAN Secretariat. https://asean.org/asean-economic-community-strategic-plan-2026-2030

ASEAN. (2025b). *ASEAN food, agriculture and forestry sectoral plan 2026–2030 (FAF-SP 2026–2030)*. ASEAN Secretariat. https://asean.org/book/asean-food-agriculture-and-forestry-sectoral-plan-2026-2030faf-sp-2026-2030/

ASEAN. (2025c). *The ASEAN Socio-Cultural Community strategic plan*. ASEAN Secretariat. https://asean.org/the-asean-socio-cultural-community-strategic-plan/

Dinas Ketahanan Pangan Provinsi Sulawesi Utara. (2025-2026). Daily commodity price reports for Bersehati and Pinasungkulan markets, Manado [Social media posts]. Facebook. https://web.facebook.com/dishanpansulut/. Data compiled manually compiled by Team Dollar Warrior (283 daily records, Jul 2025 - Jul 2026; 1,055 price observations across two markets and two commodities).

Eriyatno, E., Larasati, L., Herianto, A., Fatmaningrum, D., Bucatariu, C., & Galvez-Nogales, E. (2024). *Food loss prevention and reduction analysis in Indonesia: A case study on chili, cabbage and shallot*. Food and Agriculture Organization of the United Nations. https://doi.org/10.4060/cc8935en

Fauzia, F. A., M, A. P. D., Mahendra, M., Anggraheni, E., Haulussy, J. C., Bernier, N., & Pratama, M. A. (2023). Analisis Timbulan dan Karakteristik Komposisi Sampah di Bagian Tengah Sungai Ciliwung. *Jurnal Serambi Engineering*, 8(1). https://doi.org/10.32672/JSE.V8I1.5568

Firdaus, H., & Jaya, A. S. (2025). Scenario analysis of waste management service improvement based on projected waste generation and service capacity in Natar Subdistrict, South Lampung Regency. *Jurnal Hurriah: Jurnal Evaluasi Pendidikan dan Penelitian*, *6*(4), 1182–1191. https://doi.org/10.56806/jh.v6i4.355

Food and Agriculture Organization of the United Nations. (2016). *Post-harvest and agro-industry: Government of the Republic of the Union of Myanmar formulation and operationalization of national action plan for poverty alleviation and rural development through agriculture (NAPA): Ministry of Agriculture, Livestock and Irrigation Myanmar: National Action Plan for Agriculture (NAPA) Working Paper 13: Post-harvest and agro-industry*. https://openknowledge.fao.org/server/api/core/bitstreams/f3da75d0-9901-40ec-9623-31d81054ecf9/content

References

Food and Agriculture Organization of the United Nations. (2022). *FAO and Ministry of Agriculture start the food loss study in Indonesia*. Food and Agriculture Organization of the United Nations. https://www.fao.org/indonesia/news/detail/FAO-and-MoA-Start-the-Food-Loss-Study-in-Indonesia/en

Food and Agriculture Organization of the United Nations. (2023). *Food loss analysis causes and solutions: Chili supply chain in Banyuwangi, East Java, Indonesia: Fact sheet*. https://openknowledge.fao.org/items/d8574a63-ee69-46c9-896a-0ef409b7f63a

Food and Agriculture Organization of the United Nations. (2026). *12.3.1 Global food losses*. SDG Indicators Data Portal. https://www.fao.org/sustainable-development-goals-data-portal/data/indicators/1231-global-food-losses/en/

IPB University. (2026). Food loss and waste in horticulture reach 31.8 percent, Prof Ketty Suketi advocates for precision post-harvest management. https://www.ipb.ac.id/news/index/2026/08/food-loss-and-waste-in-horticulture-reach-318-percent-prof-ketty-suketi-advocates-for-precision-post-harvest-management

Jaroebsathapornkul, J. (2021). Value of food loss in ASEAN countries and its relationship with economic growth. *International Journal of Agricultural Technology*, *17*(1), 115–128. https://li04.tci-thaijo.org/index.php/IJAT/article/view/5914

Kementerian PPN/Bappenas. (2021). *Food loss and waste di Indonesia: Dalam rangka mendukung penerapan ekonomi sirkular dan pembangunan rendah karbon*. https://lcdi-indonesia.id/books/laporan-kajian-food-loss-and-waste-in-indonesia-dalam-rangka-mendukung-penerapan-ekonomi-sirkular-dan-pembangunan-rendah-karbon/

Krishnan, A. R., Kasim, M. M., Hamid, R., & Ghazali, M. F. (2021). A modified CRITIC method to estimate the objective weights of decision criteria. *Symmetry*, *13*(6), 973. https://doi.org/10.3390/sym13060973

Manogna, R. L., Dharmaji, V., & Sarang, S. (2025). Enhancing agricultural commodity price forecasting with deep learning. *Scientific Reports*, *15*(1). https://doi.org/10.1038/s41598-025-05103-z

NASA. (2026). *NASA POWER Data Access Viewer* [Dataset]. NASA Langley Research Center. Retrieved August 2026, from https://power.larc.nasa.gov/

Parmawati, T., Hernawan, E., & Listyarini, S. (2023). Pemodelan sistem pengelolaan sampah di tempat pemrosesan akhir (TPA) Kabupaten Tana Tidung dengan pendekatan system dynamic. *Media Ilmiah Teknik Lingkungan*, *8*(1), 17–24. https://doi.org/10.33084/mitl.v8i1.4651

References

Silva, J., Gaitán Angulo, M., Romero Borré, J., Lozano Ayarza, L. P., Pineda Lezama, O. B., del Carmen Martínez Galán, Z., & Navarro Beltran, J. (2019). An early warning method for agricultural products price spike based on artificial neural networks prediction. In *Lecture notes in computer science* (Vol. 11537, pp. 622–632). *Springer*. https://doi.org/10.1007/978-3-030-22741-8_44

United Nations. (2015a). *Sustainable Development Goal 2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture* (Target 2.4). https://sdgs.un.org/goals/goal2

United Nations. (2015b). *Sustainable Development Goal 12: Ensure sustainable consumption and production patterns (*Target 12.3). https://sdgs.un.org/goals/goal12

United Nations. (2015c). *Sustainable Development Goal 13: Take urgent action to combat climate change and its impacts* (Target 13.3). https://sdgs.un.org/goals/goal13

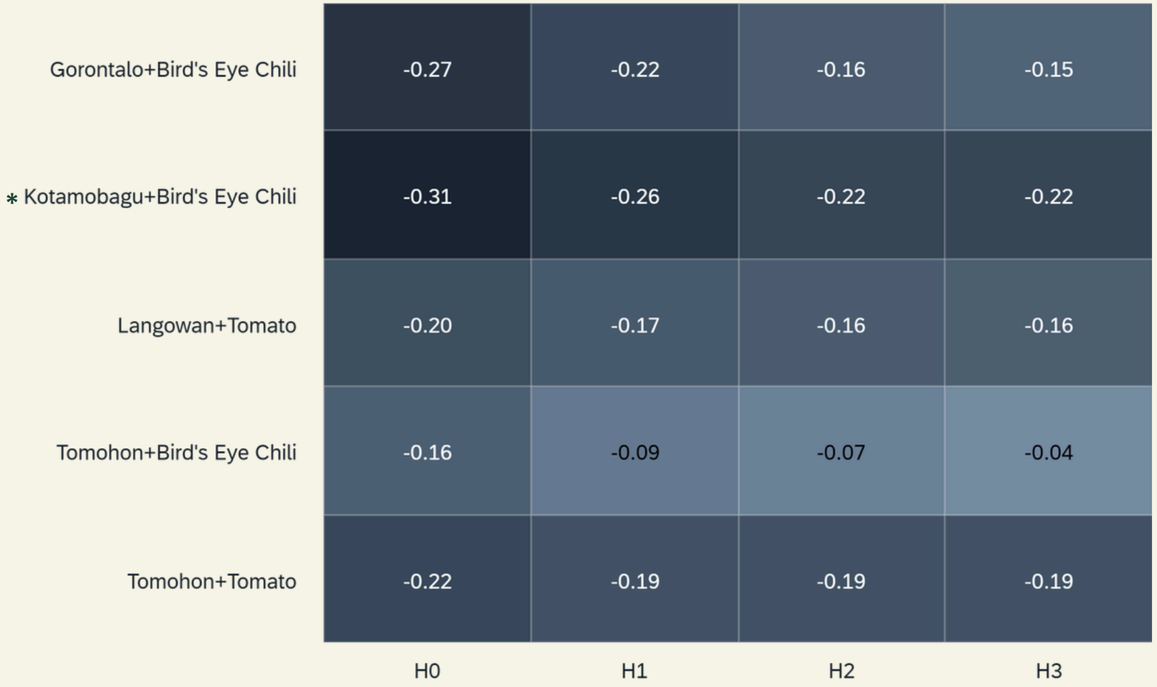
Widyawati, Rinaldi, & Winny Laura, C. H. (2020). Analisis timbulan dan komposisi sampah untuk potensi reduksi sampah di Kelurahan Selamat. *Jurnal Engineering*, *2*(2), 86-95. https://online-journal.unja.ac.id/JurnalEngineering/article/view/11507

Yang, H., Cao, Y., Shi, Y., Wu, Y., Guo, W., Fu, H., & Li, Y. (2022). The dynamic impacts of weather changes on vegetable price fluctuations in Shandong Province, China: An analysis based on VAR and TVP-VAR models. *Agronomy*, *12*(11), 2680. https://doi.org/10.3390/agronomy12112680

Appendix

Weather-Price Correlation Details

The Strength of Interregional Weather–Price Correlations and Time Lags



Key Insight

- Correlations across all 5 region-commodity pairs weaken from H0 to H3, indicating mainly short-term weather effects.
- Kotamobagu–chili shows the strongest relationship ($r = -0.31$, H0) and the only significant result ($p = 0.025$).
- Tomohon shows the weakest relationships for chili ($r = -0.16$) and tomato ($r = -0.22$), indicating limited predictive value.
- Only 1 of 20 combinations is significant, supporting the “Not Alone” argument that rainfall alone poorly explains price volatility.

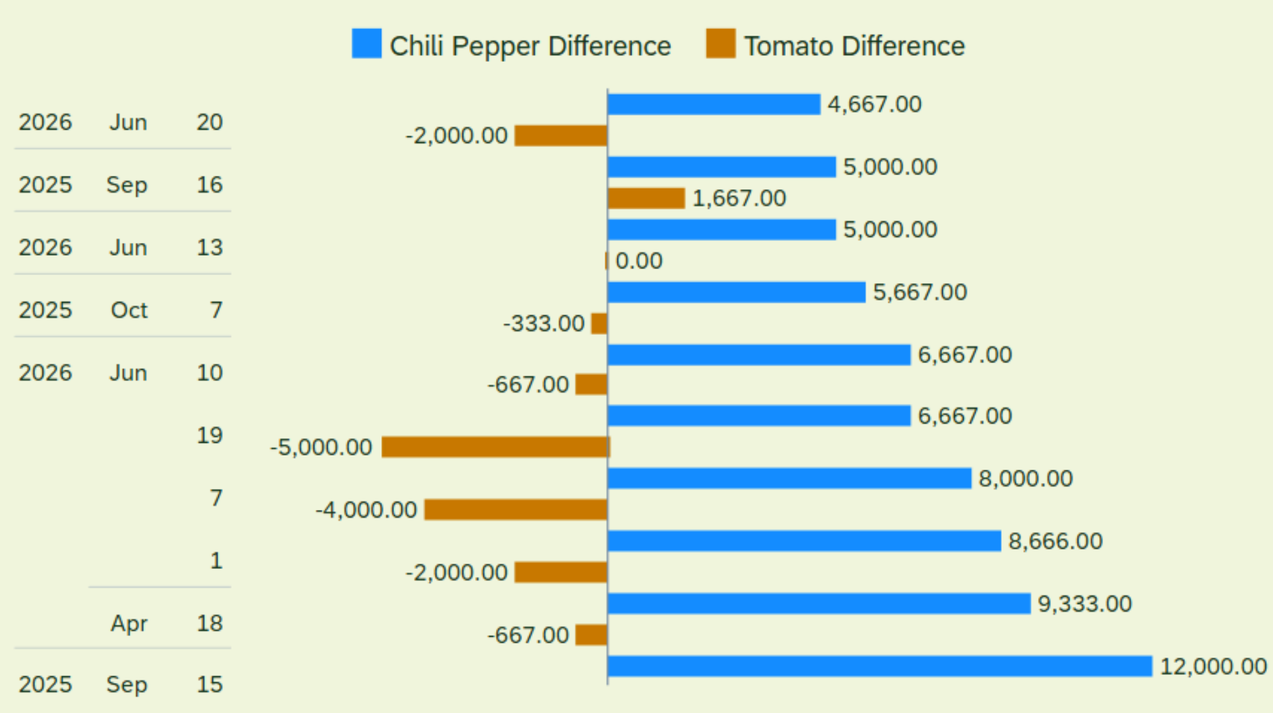
*) $p < 0.05$ is the only statistically significant correlation among the 20 combinations tested

Note: Chart generated using SAP Analytics Cloud.

Appendix

Market Price Disparity

Top 10 Days by Chili Price Disparity Chili vs. Tomato Comparison



Note: Chart generated using SAP Analytics Cloud.

Key Insight

- Chili peppers are consistently cheaper at Bersehati, while tomato prices fluctuate erratically (-3,333 to +3,000) due to weather and seasonal patterns that vary among vendors.
- Actual price disparity remains small (max -11,667) after correcting the null artifact (-58,666), supporting vendors' and market managers' view that prices are not significantly different.
- Daily granularity is important, as chili price disparity disappears when averaged monthly, indicating prices are driven by daily incoming supply volumes rather than long-term trends.
- Pinasungkulan's longer supply chain (Palu → Gorontalo → Manado) is suspected to cause greater chili price disparity compared with Bersehati's simpler supply chain.

References: Social Media Accounts of the North Sulawesi Food Security Agency from July 2025 to July 2026, Traditional Vendor Market.

Appendix

Risk Score Calculation Details

Risk Score = (W1 × **Price Trend Component**) + (W2 × **Weighted Weather Component**) + (W3 × **Price Volatility Component**)

Component 1: Price Trend (Weight: 30.5%)

Measures how quickly the current price has declined over the past several days compared with the 7-day moving average.

Price Trend Component = (MA7 Price - Current Price) / MA7 Price × 100

Logic:

If the current price is substantially lower than the 7-day moving average, the score will be higher, indicating that oversupply may be occurring. This component is assigned a high weight because price is the most directand real- time indicator of actual supply conditions.

Component 2: Source-Weighted Weather (Weight: 33.5%)

Weather Component = Σ (Rainfall at Source_i at Optimal Lag × Correlation Weight_i) / Σ Correlation Weights

Commodity	Dominant Source	Lag Used	Strength Weight (from |r|)
Chili Pepper	Gorontalo	H0 (same day)	0.27
Chili Pepper	Kotamobagu	H0 (same day)	0.31
Tomato	Tomohon	H3 (3 days prior)	0.19
Tomato	Langowan	H3 (3 days prior)	0.16

Logic:

Higher rainfall at dominant production sources at the corresponding optimal lag indicates that supply is expected to remain smooth and potentially abundant in the near term, thereby contributing to oversupply risk. This is consistent with the observed negative correlation between rainfall and price: as rainfall increases, prices tend to decline due to smoother or more abundant supply conditions.

Appendix

Oversupply Risk Score Formula

Component 3: Price Volatility (Weight: 35.9%)

Measures the standard deviation of prices over the past 7 days. Markets with high price fluctuations are more difficult for traders to predict manually, increasing the likelihood of inaccurate stock calculations.

Price Volatility Component = StdDev (Price, 7 Days) / Mean (Price, 7 Days) × 100

Note: Threshold division using 50th/75th/90th percentile-based calibration of real data (n=179). They are an initial calibration and will be validated against actual spoilage records during the pilot.

Final Risk Score Categorization

The final Oversupply Risk Score is a single number, calculated by summing the three weighted components, and then placed in one of four levels. Each level is linked to a recommended action, so traders and Perumda know what to do, not only what the score is.

Score	Level	Interpretation	Recommended Action
0 - 4.85	Low	Supply and demand are relatively balanced	Monitoring
4.86 - 9.49	Medium	Early indication of potential oversupply	Exercise caution; begin considering limited redistribution
9.50 - 13.68	High	Oversupply is highly likely to occur	Recommend active redistribution to other markets or regions
13.69 - 21.93	Critical	Severe oversupply with a high risk of food loss	Immediate action: prioritize redistribution and notify traders and Perumda

Note: "Critical" is a relative level within the observed data, not a scale out of 100.

Appendix

CRITIC Method: Risk Score Weight

CRITIC determines the weight based on contrast (standard deviation) + conflict (inter-component correlation), not on subjective assumptions.

Table 1 StdDev by Component

Component	Standard Deviation (Normalized)
Trend	0.1541
Weather	0.1674
Volatility	0.1886

Table 2 Intercomponent correlation matrix

	Trend	Weather	Volatility
Trend	1.0000	-0.1740	-0.0610
Weather	-0.1740	1.0000	-0.0860
Volatility	-0.0610	-0.0860	1.0000

Table 3 Information Content - FInal Weight

Component	Final Weight
Trend	0.305
Weather	0.335
Volatility	0.359

Note: The CRITIC weight measures the independence of information across risk components, not the predictive power of weather on prices (see the separate page on weather-price correlations, r = -0.27).

References: Krishnan el al. (2021).

Appendix

Cost Estimation Breakdown

TABLE 1 HARDWARE CAPEX

No	Hardware Component	Technical Specifications for Goods/Services	Volume	Estimated Unit Price (IDR)
	Phase 1: AI Pilot Infrastructure			
1.	IP CCTV Camera	Low: Hikvision 4MP IP CCTV Outdoor IP67 PoE, High Hikvision 8MP/4K ColorVu IP CCTV PoE	1 unit (Low), 2 unit (High)	1,200,000 - 2,250,000
2.	Edge Computing Device	Low: NVIDIA Jetson Nano Dev Kit (4GB RAM), High: NVIDIA Jetson Orin Nano Dev Kit (8GB RAM)	1 unit	2,500,000 - 6.500,000
3.	Accessories & Local Networks	Outdoor Cat6 LAN Cable, RJ45, PoE Injector, Gigabit PoE Switch, Weatherproof Metal Panel Box, Bracket	1 Package	Variatif
4.	Physical Installation Service	CCTV installation, conduit pulling, panel mounting, mini UPS power setup	1 Package	Lumpsum
	Phase 2: IoT Sensor Expansion			
1.	Node Sensor Los Pasar	ESP32 Microcontroller + DHT22/AM2301 Temperature & Humidity Sensor + IP65 Enclosure & Lithium Battery	30 Unit (Low), 60 Unit (High)	150,000 - 200,000
2.	Receiving IoT Gateway	Single-Board Computer (Raspberry Pi 4) + Receiver Module + 4G LTE Cellular Module	1 Unit (Low), 2 Unit (High)	1,500,000
3.	Data Communication Setup	Node assembly services, wireless MQTT protocol programming, battery transmission testing	1 Package	Lumpsum
4.	Installation & Calibration	Installation of nodes at 30–60 pilot vendors; calibration of sensors to the ambient temperature of the wet market	1 Package	Lumpsum

Appendix

Cost Estimation Breakdown

TABLE 2 SOFTWARE CAPEX

No	Software Modules & Features	Development Team Work Details	Total for the Low-End Scenario (IDR)	Total for the High-End Scenario (IDR)
1	YOLOv8 Detection Model (Layer 2)	Low: Basic YOLOv8 transfer learning model using a public dataset. High: Custom dataset collection (photos of local chili peppers and tomatoes from Pasar Bersehati), manual image annotation, custom training and fine-tuning.	6,000,000.00	15,000,000.00
2	Receipt Reader OCR Module (Layer 3)	Low: Scanner for printed inventory receipts using a standard open-source OCR (PaddleOCR).High: Advanced layout OCR engine for handwritten receipts from market vendors.	4,000,000.00	10,000,000.00
3	Visualization & Workflow Dashboard	Low: Functional web dashboard (Streamlit / Flask) with a local SQLite database. High: Enterprise-level React.js & Node.js, secure multi-user login, integration for sending automated WhatsApp push notifications to administrators.	5,000,000.00	12,000,000.00
4	Prediction & Enterprise API Integration	Low: Integration of basic historical price statistics and public weather data APIs. High: Development of APIs for integrating circular logistics with SAP Integrated Business Planning (IBP) and SAP Analytics Cloud (SAC).	3,000,000.00	9,000,000.00
	SUBTOTAL SOFTWARE CAPEX	*Wages for Integrated Software Development Work.*	*18,000,000.00*	*46,000,000.00*

Appendix

Cost Estimation Breakdown

TABLE 3 OPEX / Year

No	Operating Expense Account	System Requirements Description	Total for the Low-End Scenario (IDR/Year)	Total for the High-End Scenario (IDR/Year)
1	Cloud Server & Database Rental	Low: Lightweight Cloud VPS GPU server for hosting databases and dashboards. High: Industry-grade managed Cloud GPU AI server with automatic backups.	4,200,000.00	14,400,000.00
2	Internet Data Plans by Network	Low: Monthly IoT SIM card for data transmission from Jetson to the cloud. High: 24-hour non-stop Industrial Dual-SIM LTE Router Subscription.	3,000,000.00	6,000,000.00
3	Security & API Licenses	System-specific domain, premium SSL HTTPS, paid high-resolution weather forecast API license.	1,100,000.00	3,400,000.00
4	Field Maintenance & Calibration	Low: Routine remote maintenance and remote bug fixes. High: Monthly on-site visits, cleaning camera lenses of market dust, replacing IoT sensor batteries, and periodic AI model retraining.	3,000,000.00	6,000,000.00
	SUBTOTAL OPEX	*Net Operating Expenses*	*11,300,000.00*	*29,800,000.00*
	Contingency Fund	Anticipate annual cost fluctuations	1,700,000	4,470,000
	TOTAL OPEX / YEAR	Inclusive Contingency	13,000,000	34,270,000

Appendix

Loss Calculation Details

1. Raw Field Data - Bersehati Market

Metric	Value	Source
Barito's stall is full	207	Perumda Pasar Manado web
Barito's stall is empty	5	Perumda Pasar Manado web
Total capacity of Barito stalls	212	Perumda Pasar Manado web

Note: The number 207 refers to a combined stall selling Barito commodities (onions, chili peppers, and tomatoes) and is not a stall exclusively forchili peppers and tomatoes.

2. Key Assumptions

Since the figure of 207 combines three commodities, we applied an explicit assumption to isolate the share of chili peppers and tomatoes: chili peppers and tomatoes are estimated to account for 60–70% of the stalls at Barito Market

The median estimate of 65% was used:

NBersehati = 207 x 0.65 ~ 135 vendor

3. Calculation Formula & Parameter Validation

VLB = N x VLH X 365 day

VS = VLB x R

Variable	Description	Value	Source
VLH	Daily Volume Loss per Trader (kg/day), i.e., the average of wasted/unsold commodities	1.04 kg/day	Obtained directly through empirical data based on the average of six field interview results with chili and tomato vendors at Bersehati Market. This figure is supported by research by Widyawati et al. (2020), which found that the average waste generation at traditional markets reaches 2.74 kg per person per day, with market waste consisting predominantly of 94.1% organic waste.
NBersehati	Number of active chili pepper & tomato traders per market	~135 vendor	207 active Barito stalls × 65% spatial commodity assumption.
R	Targeted reduction rate of the solution, i.e., the percentage saved from early warning + redistribution	15% / 20%/ 30%	In accordance with technical feasibility and policy limits. Widyawati et al. (2020) demonstrated that the potential for reducing organic market waste through composting reaches 37.33%. On the other hand, Parmawati et al. (2023), in their waste management policy modeling, established regional waste reduction scenarios of 8% (pessimistic), 33.73% (moderate), and 48.336% (optimistic).

Appendix

Loss Calculation Details

4. Result - Bersehati Annual Loss Volume

VLBBersehati = 135x 1.04/kg/day
= 51,246 kg/year
~ 51.2 tons/year

Macro-Level Perspective: A study by Fauzia et al. (2023) shows that the average waste generation at a single traditionalmarketis 1.05 metric tons per day.Based on this data, the accumulated food loss of 51.2 metric tons per year at Bersehati Market is equivalent to halting the disposal of the entire waste generated by a single traditional market for 49 consecutive days.

5. Volume Saved

Scenario	Reduction Rate (R)	Volume Saved (VS)	Landfill Area Saved
Conservative	15%	7,687 kg/year (7.7 tons)	26.1 m^3 / year
Moderate	20%	10,249 kg/year (10.2 tons)	34.7 m^3/year
Optimistic	30%	15,374 kg/year (15.4 tons)	52.1 m^3/year

Justification for Landfill Volume Conversion: Fauzia et al. (2023) (citing Tchobanoglous et al., 1993) state that municipal solid waste density ranges from 175 to 415 kg/m^3, with a median value of 295 kg/m^3.Using this density conversion factor, our moderate scenario saves 34.7 m^3 of landfill space each year.

6. Economic Value Estimate (Moderate Scenario)

10,249 kg xRp57,000/kg
= ~ Rp 584 million/year

The average price of red chili peppers, at Rp 57,000 per kilogram, is based on an analysis of 12 months of historical price trends in the local market.

Appendix

Frequently Asked Questions

1. Why use a "Three-Layer AI" approach instead of a single predictive model?

Traditional markets are complex and unpredictable, so relying on time-series data alone isn't enough to reflect real conditions on the ground. Our system therefore combines three approaches: time-series models for forecasting, YOLOv8-based computer vision for visual ground-truth data, and OCR to digitize manual records. This way, predictions are backed not just by historical data, but also by real-world field validation.

2. Is this solution scalable to other ASEAN countries?

Yes, the architecture is highly modular. The YOLOv8 model can be retrained to recognize different local commodities, and the OCR module works broadly since manual record-keeping is common across ASEAN traditional markets. This makes the system well-suited for regional replication under the ASEAN Integrated Food Security (AIFS) Framework.

3. How do you mitigate the risk of prediction errors

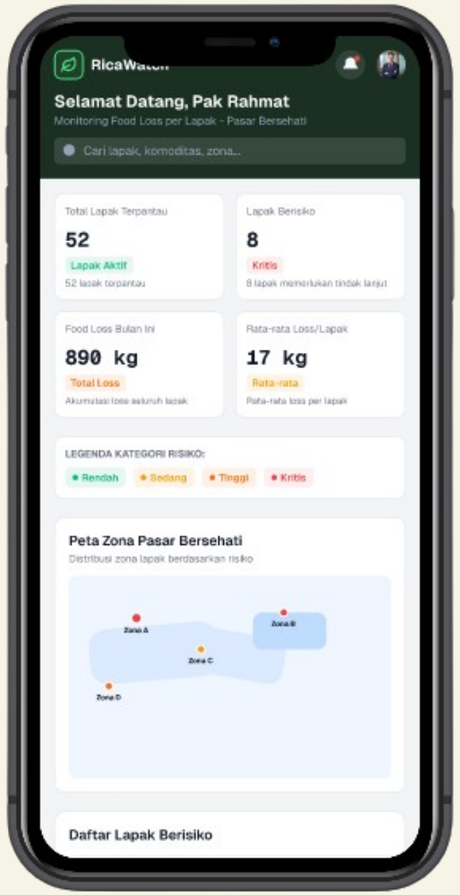
Weather-price correlations tend to be weak when applied at a national level. Our model addresses this by calculating correlations per supply source instead of relying on a national average, which prevents errors from being "averaged out." This allows us to generate a more granular Risk Matrix (Low to Critical), helping traders take preventive action rather than depend on generic, one-size-fits-all statistics.

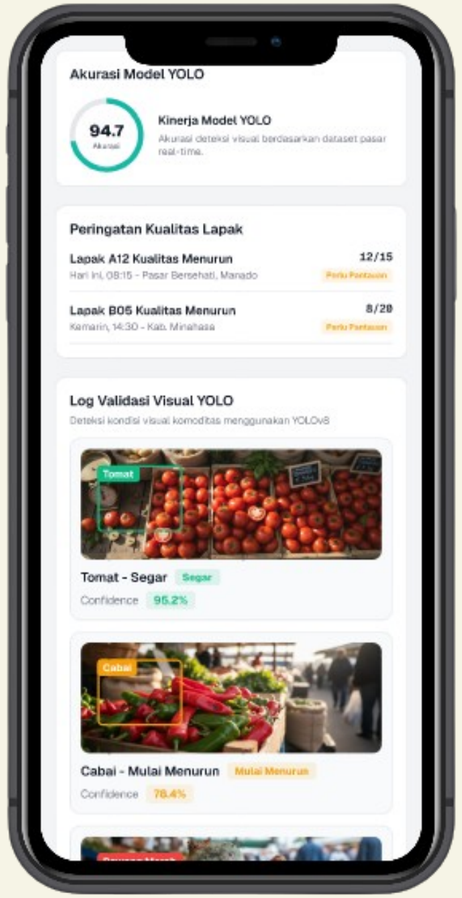
4. How do you ensure data integrity from inconsistent market sources?

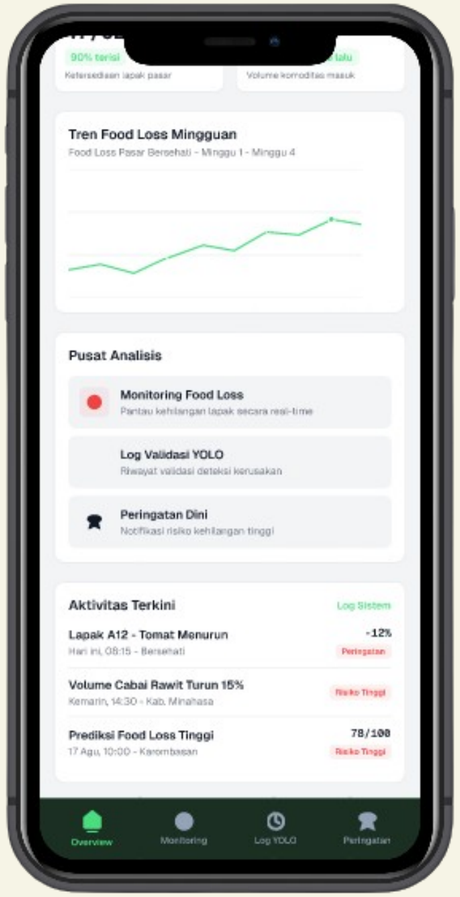
We maintain strict data hygiene: small gaps (1–3 days) are filled through interpolation, while larger gaps (more than 7–10 days) are left blank to preserve the validity of the analysis. We further cross-check these results with field interviews, ensuring our insights are grounded in real-world conditions rather than raw data alone.

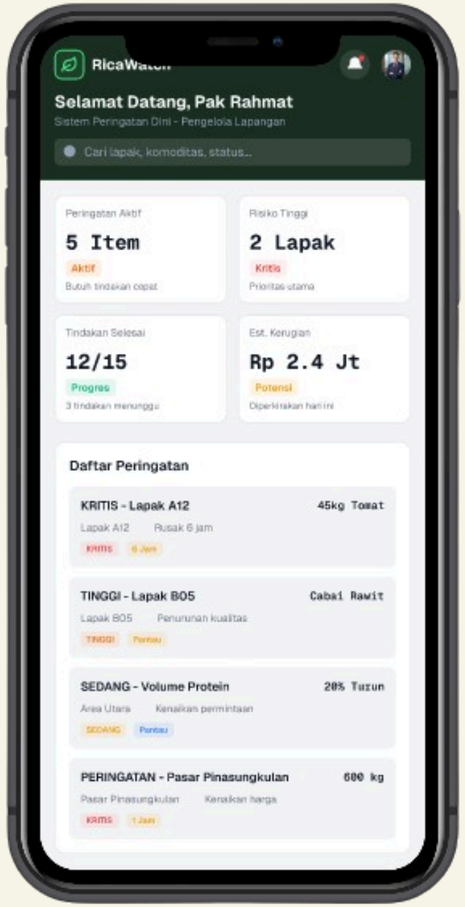
Appendix

Application Mockup (Manager)









Appendix

Application Mockup (Vendor)

